

Data Analysis Prelim Exam

Department of Statistics, University of Chicago

Thursday, Sept 19, 2019: Due Friday, Sept 20 at 4.00pm

1 Data

The data set for this exam is taken from the paper

Effects of climate warming on photosynthesis in boreal tree species depend on soil moisture, Reich et al, *Nature* vol. 562, pages 263–267 (2018)

The data are provided in the file `borealwarming.csv`.

Experiment design The paper studies the effect of warmer temperatures on photosynthesis in trees. Two sites a large distance apart were split into three blocks each, and each block was split into four plots. In each plot, warming cables were laid in the soil with the capability of maintaining a temperature that is 3.4 degrees Celsius warmer than the ambient temperature. Each plot was then randomly assigned to one of two treatments—**warmed**, where the underground cables and above-ground heating lamps were turned on during the growing season, or **ambient**, where the electrical cables were not turned on. Within each plot, trees from 11 species were planted and measurements were taken for the photosynthesis activity level (measured in a single leaf). The soil water level was also measured.

Variables The principal variables are:

- **site**, **block**, **plot_id** denote the location of the plant being measured. There are two sites, six blocks (three per site), and 24 plots (four plots per block).
- **warming_treatment** takes values **warmed** or **ambient**, for the treatment or control group. The randomization is at the level of the plot (i.e., all trees in the same plot are assigned the same treatment).
- **species** takes 11 values. Four species (**abiba**, **picgl**, **pinba**, **pinst**) are needle leaved (coniferous) trees, while seven species (**aceru**, **acesa**, **betpa**, **poptr**, **quema**, **queru**, **rhaca**) are broadleaf (deciduous) trees.
- **year** is the year when the measurement is taken (2009, 2010, or 2011).
- **doy** is the day of the year (e.g., **doy**=1 means that the measurement was taken on January 1st of that year). Note that two measurements may have the same **doy** value even if they occur in different years.
- **Asat** measures the photosynthesis activity of the leaf; 2049 measurements.
- **soil_water_VWC** measures the soil water content in the plot. (On any given day, this measurement is the same for all plants in the same plot.)
- You may ignore the other variables present in the data set.

2 Questions

You are encouraged to read the *Nature* paper before answering the questions, and you are not restricted to the specific questions mentioned below; you are free to examine other points of interest if you like. Throughout, you may set aside missing response values, and you may assume that measurements are independent across different days.

Your report should include code for all plots and results that you present.

1. Before any trees were planted, a grid of underground electrical cables was laid down at a depth of 15 cm in each plot, with sufficiently small separation that the heating effect could be deemed uniform. The cables in the treated plots were used as heating elements. What was the purpose of the cables in the control plots?
2. Use the data to reproduce the authors' plots in Fig. 1 in a similar format. What, if anything, do these plots tell you about the effect of elevated temperature on the various species? (Note that the species can be grouped into deciduous vs. coniferous trees as given above.)
3. Soil water content is expected to vary from day to day with the most recent weather and from plot to plot depending on the topography, for example, plot elevation, exposure, drainage capacity of the sub-soil, and so on. For this question, examine the effect of the treatment assignment (**warming** vs. **ambient**) on the soil water content (**soil_water_VWC**), taking appropriate account of such variations.
 - (a) First, run an analysis assuming that the treatment effect is constant over sites and over years. You should fit a suitable linear Gaussian model with soil water content – or some suitable transformation thereof – as response. Explain how you fitted the model, show the parameter estimates with standard errors, and discuss conclusions about the effect of the treatment.
 - (b) You should have found a small negative treatment effect in the first part of this question. Next, examine the question of whether the treatment effect is indeed constant across sites, as assumed in the first part of the question. Similarly, examine whether it is constant across years. Explain how you might address these questions, and report your answers.
4. For this question we will use the data from only the **queru** species (northern red oak). The photosynthesis level of a tree may depend on both the soil water content and the treatment (in addition to variation from day-to-day weather and effects of other factors).
 - (a) Build a linear Gaussian model with photosynthesis (**Asat**) – or some suitable transformation thereof – as the response. Your analysis should accommodate block, plot, and/or temporal effects as needed. Explain how you fitted the model, and show the parameter estimates with standard errors. Give a brief summary of the conclusions reached on the basis of the models fitted, and report a conclusion testing the following hypothesis with a likelihood ratio test: the effect (slope) of soil water content on photosynthesis level is the same for the two treatment groups.
 - (b) Imagine that in the year 2010 you measured **Asat** on a red oak leaf in a new plot at site **hwrc** that was assigned to the warming treatment. Under the model you estimated what would be the mean and SD of that measurement? Provide an SE for these quantities. Show code for your calculations, and explain how you arrived at your conclusions.
5. Summarize the assumptions made in the models you fitted in questions 3 and 4, and discuss the validity and implications of these assumptions.
6. Time permitting: Let Y be a Gaussian random vector with mean $X\beta$, covariance matrix $\Sigma = W^{-1}$, and let $R = QY$ be the residual projection in which $Q = I - X(X'WX)^{-1}X'W$. Compute the mean and variance of the residual quadratic form $R'\Delta R$ in which Δ is a given symmetric matrix. Your answers should be expressed in terms of the matrices Q, Σ, Δ .
 Let Δ be the indicator function for consecutive calendar days, and let Y be the soil water content. Compare the observed residual quadratic form with its mean and variance as computed for the fitted model in part 3(b) with two interactions. What does this calculation suggest?